

COLLECTION PROPERTIES

Property	List	Tuple	Dictionary	Set	String
Data Type	Mutable	Immutable	Mutable	Mutable	Immutable
Syntax	[]	()	{ key: value, ... }	{ }	'', ""
Ordered	Yes	Yes	No	No	Yes
Duplicate Values	Allowed	Allowed	Keys: Not allowed, Values: Allowed	Not allowed	Allowed
Access by Index	Yes	Yes	No (Access by key)	No	Yes
Modification	Add, Remove, Update	Not applicable	Add, Remove, Update	Add, Remove, Update	Not applicable
Iteration	Yes	Yes	By key, value, or items	Yes	Yes
Common Operations	append(), remove(), pop(), insert(), extend()	Not applicable	get(), update(), pop(), del, keys(), values(), items()	add(), remove(), discard(), pop(), clear(), update(), set operations	String methods: split(), join(), find(),
Use Cases	Storing sequences of items where order matters and	Storing sequences of items where order matters and duplicates are	Mapping keys to values for quick lookup.	Storing unique elements where order doesn't matter and	Storing and manipulating text data.

	duplicates are allowed.	allowed but need immutability		quick membership tests are needed.	
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Lists

- **Definition:** Ordered, mutable collections of items.
- **Use Cases:** When you need an ordered collection that can be modified (e.g., adding, removing, or changing elements).
- **Example Scenarios:**
 - Storing a sequence of items where order matters (e.g., a to-do list).
 - Implementing stack or queue structures.
 - Collecting items dynamically during runtime.

```
# Example: Task management
tasks = ["task1", "task2", "task3"]
tasks.append("task4") # Adding a new task
tasks.remove("task2") # Removing a task
```

Tuples

- **Definition:** Ordered, immutable collections of items.
- **Use Cases:** When you need an ordered collection that should not change after its creation, often used for fixed collections of items.
- **Example Scenarios:**
 - Storing a fixed set of values, like coordinates or RGB color values.
 - Returning multiple values from a function.
 - Using as keys in dictionaries (since tuples are hashable and immutable).

```
# Example: Storing coordinates
point = (10, 20)
```

```
# Example: Returning multiple values from a function
def get_min_max(numbers):
    return (min(numbers), max(numbers))
```

Sets

- **Definition:** Unordered collections of unique items.
- **Use Cases:** When you need to store unique items and perform operations like union, intersection, and difference efficiently.
- **Example Scenarios:**
 - Removing duplicates from a list.
 - Checking membership quickly.
 - Performing set operations (union, intersection, difference).

```
# Example: Removing duplicates
items = [1, 2, 3, 1, 2, 4]
unique_items = set(items) # {1, 2, 3, 4}
```

```
# Example: Set operations
set1 = {1, 2, 3}
set2 = {3, 4, 5}
union_set = set1 | set2    # {1, 2, 3, 4, 5}
intersection_set = set1 & set2 # {3}
```

Dictionaries

- **Definition:** Unordered collections of key-value pairs.
- **Use Cases:** When you need to associate values with unique keys for fast lookup, addition, and deletion of key-value pairs.
- **Example Scenarios:**
 - Storing and retrieving data using unique identifiers (e.g., a phone book).
 - Counting occurrences of items.
 - Representing complex data structures (e.g., JSON-like data).

```
# Example: Phone book
phone_book = {"Alice": "123-4567", "Bob": "987-6543"}
phone_book["Charlie"] = "555-0000" # Adding a new entry
```

```
# Example: Counting occurrences
items = ['apple', 'banana', 'apple', 'orange']
counts = {}
```

```
for item in items:
    if item in counts:
        counts[item] += 1
    else:
        counts[item] = 1
```

Summary

- **List:** Use when you need an ordered, mutable collection. Example: A list of tasks to be done.
- **Tuple:** Use when you need an ordered, immutable collection. Example: Storing a set of coordinates.
- **Set:** Use when you need an unordered collection of unique items. Example: Removing duplicates from a list of items.
- **Dictionary:** Use when you need to store key-value pairs for fast lookups. Example: A phone book mapping names to phone numbers.